

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester II Examination, 2024-25 (NEP)

CHEMISTRY**Paper Code: CHEMJ21 / CHEMN21 : Basic Concepts of Chemistry-II**

Ref. No.: CECNF/N/2024-25/10 & 11

Time: 3 Hours

Maximum Marks: 70

Instructions: Attempt five questions including Question 1 which is compulsory. Attempt at least one question from Questions 2 and 3. The figures in the right-hand margin indicate marks.

Question 1. Answer all parts:

- (a) "Carbon possesses only two unpaired spins in its ground state yet it forms four bonds." Why? [2]
- (b) Comment upon stability of Be_2 as per molecular orbital theory. [2]
- (c) "The HCF bond angle is slightly smaller than HCH in CH_3F ." Explain this observation in terms of Bent's rule. [1]
- (d) Explain why the phenols are stronger acids than alcohols but weaker than carboxylic acids. [3]
- (e) Why fully-eclipsed Newman projection of n-butane have maximum Torsional Strain? [2]
- (f) Why does only the E2 elimination show the isotope effect and E1 does not? [1]
- (g) In Williamson's ether synthesis, why primary alkyl halide used instead of a secondary or tertiary one? [2]
- (h) How many transition states exist in the $\text{S}_{\text{N}}1$ reaction? [1]

Question 2.

- (a) Explain in brief the Valence bond theory of metallic bonding through a suitable example. [2]
- (b) "The boiling point of HF is much higher than that of HCl, HBr and HI while in aqueous state it is poorest acid among all." Why? [4]
- (c) Discuss the structure and bonding of either Methyl Lithium or diborane. [4]
- (d) Draw the crystal structure of Rock salt and mention the co-ordination nos. of sodium and chloride ions. [2]
- (e) Why is Si-F bond stronger than corresponding bond with Carbon? [2]

Question 3.

- (a) Let A and B be two atoms with wave functions for 1s orbitals as ψ_A and ψ_B respectively. Write the molecular orbital wave functions using these atomic orbital wave functions

ψ_A and ψ_B . Write expressions using molecular orbitals for the probability of finding the electron density in a unit volume and interpret the meaning of each term in the expressions. [5]

- (b) "The compounds trimethylamine $(\text{CH}_3)_3\text{N}$ and Trimethylsilylamine $(\text{Me}_3\text{Si})_3\text{N}$ have similar formulae but have pyramidal and triangular planar structures respectively." Why? [3]
- (c) Explain in brief the molecular orbital theory of metallic bonding through a suitable example. [2]
- (d) Comment upon variation of bond angles in CH_4 , NH_3 and H_2O in the light of VSEPR theory. [2]
- (e) Explain the defects noted in extending homonuclear diatomic molecular orbital concepts in the heteronuclear diatomic molecule CO using energy level diagram. [2]

Question 4.

- (a) Write down the product(s) of the following reactions: [8]
- (i) Cyclopentadiene + Maleic anhydride $\xrightarrow{0^\circ\text{C}}$?
- (ii) $\text{H}_3\text{C} - \text{C} \equiv \text{CH} \xrightarrow{\text{dil. H}_2\text{SO}_4/\text{HgSO}_4}$?
- (iii) Cyclopentene $\xrightarrow{1. \text{MCPBA}/2. \text{H}^+}$?
- (iv) $n\text{H}_3\text{C} - \text{CH} = \text{CH}_2 \xrightarrow{\text{Benzoyl peroxide}/\Delta}$?
- (b) Arrange the following in increasing order of basicity with a suitable explanation: [4]
- (i) Ph-NH_2
- (ii) Ph_2NH
- (iii) Ph_3N
- (c) Predict the E and Z nomenclature of the following: [2]
- (i) 1-bromo-1-chloropropene
- (ii) 2-bromo-1-chloropropene

Question 5.

- (a) Write down the major products formed in the following reaction with a suitable mechanistic explanation: [8]
- $\text{CH}_3 - \text{CH}_2 - \text{C}(\text{CH}_3)(\text{Br}) - \text{CH}_3 \xrightarrow{\text{KOH}/\text{EtOH}}$ Substitution (?) / Elimination (?)
- (b) What is the intermediate involved in the E1cB mechanism? [2]
- (c) Identify the products formed in the following reaction with a suitable mechanism: [4]
- $\text{CH}_3 - \text{O} - \text{C}(\text{CH}_3)_3 + \text{HI} \rightarrow ?$

Question 6.

- (a) Predict the R and S nomenclature of glyceraldehyde and threonine derivatives. [3]
- (b) How do you prepare the following from phenylmagnesium iodide? [6]

- (i) Benzenesulfonic acid (Ph-SO₂H)
- (ii) Phenol (Ph-OH)
- (iii) Benzyl alcohol (Ph-CH₂OH)
- (c) How do you prepare the following from ethyl acetoacetate? [5]
 - (i) 1,4-diketone
 - (ii) 1,6-diketone

Question 7.

- (a) Identify the products formed in the following reaction with a suitable mechanism: [4]
 $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2 \xrightarrow{\text{Br}_2} ?$
- (b) How do you prepare the following from diethyl malonate? [4]
 - (i) Glutaric acid
 - (ii) Dimethyl acetic acid
- (c) Discuss the S_Ni reaction with a suitable example. [4]
- (d) Distinguish between primary and secondary alcohols by the substitution method. [2]
